

FEDERAL BOARD OF INTERMEDIATE & SECONDARY EDUCATION

Annual Examination 2026 — ANSWER KEY (Version 1)

Physics — Class 9 (Science Group)

Syllabus: Unit 1 — Physical Quantities & Measurement | Unit 2 — Kinematics | Unit 3 — Dynamics-I | Unit 4 — Dynamics-II | Unit 5 — Pressure & Deformation in Solids | Unit 6 — Work & Energy (till 6.4) | Unit 9 — Nature of Science and Physics

Total Marks: 46

Time: 2 Hours

Paper: Exam-15 (Version 1)

SECTION A — Multiple Choice Questions ($30 \times 1 = 30$ Marks)

Q1. The number of significant figures in 0.02080 kg is: → **C – 4** 1 mark

Leading zeros (before the first non-zero digit) are **never** significant. The digits 2, 0, 8, 0 are all significant: the zero between 2 and 8 is **between** significant digits, and the final trailing zero comes **after the decimal point**, so both count. Total = 4 significant figures.

SLO: U1 §1.10, P-09-A-16 (rounding/significant-figure conventions). Source: U1 p.30–31, general rules for writing significant figures.

Q2. Arrows land in a tight cluster, far from the bullseye: → **B – Precise but not accurate** 1 mark

Precision is the consistency/repeatability of measurements (how close they are to each other); **accuracy** is how close a measurement is to the true value. The arrows are tightly grouped (precise) but off-target (not accurate).

SLO: U1 §1.9, P-11-A-09 (differentiate precision and accuracy).

Q3. Instrument to measure WEIGHT directly: → **B – Spring balance (force meter)** 1 mark

A **physical balance** compares an unknown mass against standard weights and reads **mass**. A **spring balance** uses the extension of a spring (Hooke's law) and its scale is calibrated directly in newtons, so it reads **weight** (a force).

SLO: U1 §1.7.4 (physical balance); U3 P-09-B-22 (force meter measures weight); U5 §5.4.1 (spring scale).

Q4. Water rises from 42 mL to 65 mL: → **A – 23 cm³** 1 mark

Volume of stone = final reading – initial reading = $65 - 42 = 23 \text{ mL} = 23 \text{ cm}^3$ (the stone displaces its own volume of water).

SLO: U1 §1.7, P-09-A-13 (measuring cylinder to measure volume).

Q5. Ball dropped, reaches 35 m/s: → **C – 3.5 s** 1 mark

$v = u + gt$, with $u = 0$. So $t = v/g = 35 / 10 = 3.5 \text{ s}$.

SLO: U2, P-09-B-10 (free-fall acceleration to solve problems).

Q6. Horizontal line on a distance-time graph means: → **A – At rest** 1 mark

The distance from the reference point is not changing with time, so the body is **stationary (at rest)**.

SLO: U2, P-09-B-9 / P-09-B-11 (sketch/interpret distance-time graphs; gradient gives speed).

Q7. Train's speed 10→40 m/s in 15 s, GRADIENT represents: → **A – Acceleration, equal to 2 m/s²** 1 mark

Gradient of a speed-time graph = $\Delta v / \Delta t = (40 - 10) / 15 = 2 \text{ m/s}^2$, the train's acceleration. (The gradient never gives distance or displacement — that is the AREA.)

SLO: U2, P-09-B-14 / P-09-B-15 (gradient of speed-time graph gives acceleration).

Q8. Car at constant 12 m/s for 25 s, AREA represents: → **B – Distance travelled, equal to 300 m** 1 mark

Area under a speed-time graph = distance. For constant speed the shape is a rectangle: Area = speed $\times$ time = $12 \times 25 = 300 \text{ m}$. (The area never gives acceleration — that is the GRADIENT.)

SLO: U2, P-09-B-12 / P-09-B-13 (area beneath speed-time graph gives distance).

Q9. Trolley 0.5 kg, 20 m/s, stopped in 0.05 s. Impulse: → **D – 10 N·s** 1 mark

Impulse = change in momentum = $m(v - u) = 0.5 \times (0 - 20) = -10 \text{ kg·m/s}$, magnitude **10 N·s**. (Option B, 200 N·s, is the common mistake of calculating the average *force* $F = \Delta p / \Delta t = 10 / 0.05 = 200 \text{ N}$ instead of the impulse itself — impulse and force are not the same thing.)

SLO: U3 §3.6, P-09-B-40 (define and calculate impulse).

Q10. Bullet 0.01 kg/200 m/s embeds in 1.99 kg block: → **A – 1 m/s** 1 mark

Total momentum before = $(0.01)(200) + (1.99)(0) = 2 \text{ kg·m/s}$. After impact the combined mass moves together: total mass = $0.01 + 1.99 = 2 \text{ kg}$. By conservation of momentum, $v = 2 / 2 = 1 \text{ m/s}$.

SLO: U3, P-09-B-44 (apply conservation of momentum in one dimension).

Q11. $F = 18 \text{ N}$ produces $a = 2 \text{ m/s}^2$. Mass: → **C – 9 kg** 1 mark

From $F = ma$, $m = F/a = 18 / 2 = 9 \text{ kg}$.

SLO: U3, P-09-B-30 (state and apply Newton's second law).

Q12. Table 3.1, proton-electron interaction is an example of: → **B – Electromagnetic force** **1 mark**

Table 3.1 (Fundamental Forces in Nature) shows the proton–electron interaction, mediated by photons, as the textbook's example of the **electromagnetic force** (it acts between electric charges).

SLO: U3 §3.1.2, Table 3.1; P-09-B-25 (three/four fundamental forces and their relative strengths).

Q13. Seesaw: 30 N at 2 m (left); balancing weight at 3 m (right)? → **A – 20 N** **1 mark**

By the principle of moments, clockwise moment = anticlockwise moment: $30 \times 2 = W \times 3$, so $W = 60/3 = 20 \text{ N}$.

SLO: U4, P-09-B-45 / P-09-B-46 (equilibrium using the principle of moments).

Q14. Four engines, all thrust in the same direction: → **D – Like parallel forces** **1 mark**

Parallel forces acting in the **same direction** are called like parallel forces (e.g. pushing a cart with both hands, or here, all four engines thrusting forward together).

SLO: U4 §4.1, P-09-B-43 (differentiate like and unlike parallel forces).

Q15. Skydiver reaches terminal velocity when: → **C – Her weight equals the air resistance (net force is zero)** **1 mark**

Terminal velocity is reached when the upward air resistance exactly balances the downward weight, so the net force (and hence acceleration) becomes zero and the body falls at a constant maximum speed.

SLO: U4, P-09-B-36 (analyse the dynamics of an object reaching terminal velocity).

Q16. Disadvantage of friction: → **B – Causes wear and tear of moving parts** **1 mark**

The textbook lists walking, driving, and holding nails/screws in wood as **advantages** of friction; wear and tear (along with energy loss as heat) is listed as a **disadvantage**.

SLO: U4 §4.6.2 (advantages and disadvantages of friction).

Q17. Well-type manometer, $\rho=1200 \text{ kg/m}^3$, $h=15 \text{ cm}$. Pressure difference: → **D – 1800 Pa** **1 mark**

$\Delta P = \rho gh = 1200 \times 10 \times 0.15 = 1800 \text{ Pa}$.

SLO: U5 §5.5, P-09-B-85 (workings and applications of a manometer).

Q18. 4 kg hung from spring, extends 2 cm. Spring constant: → **C – 2000 N/m** **1 mark**

Force stretching spring = weight = $mg = 4 \times 10 = 40 \text{ N}$. By Hooke's law, $k = F/x = 40 / 0.02 = 2000 \text{ N/m}$.

SLO: U5 §5.2, P-09-B-56 (define and calculate the spring constant).

Q19. One pascal is defined as a pressure of: → **A – 1 N/m²** **1 mark**

Pressure = Force / Area, so its SI unit, the pascal, is simply **1 N/m²** (one newton of force acting on one square metre).

SLO: U5, P-09-B-77 (define and calculate pressure).

Q20. Hydraulic press: $A_1=0.0015 \text{ m}^2$, $A_2=0.45 \text{ m}^2$, $F_1=20 \text{ N}$. Output force: → **D – 6000 N** **1 mark**

By Pascal's principle, pressure transmitted equally: $F_2 = F_1 \times (A_2/A_1) = 20 \times (0.45/0.0015) = 20 \times 300 = 6000 \text{ N}$.

SLO: U5 §5.6, P-09-B-86 (define and apply Pascal's law).

Q21. Three common types of manometer: → **B – U-tube, Well Reservoir, and Inclined manometers** **1 mark**

The textbook names and illustrates exactly these three designs: (a) U-tube Manometer, (b) Well Reservoir Manometer, (c) Inclined Manometer.

SLO: U5 §5.5, P-09-B-85.

Q22. 4 kg ball dropped from 10 m. Kinetic energy at impact: → **C – 400 J** **1 mark**

PE at the top = $mgh = 4 \times 10 \times 10 = 400 \text{ J}$. By conservation of energy (no air resistance), all of this PE converts to KE just before impact: KE = **400 J**.

SLO: U6, P-09-B-64 / P-09-B-65 / P-09-B-68 (KE, PE, conservation of energy).

Q23. Sledge pulled with 90 N at 60°, moved 3 m. Work done: → **A – 135 J** **1 mark**

$W = Fd \cos\theta = 90 \times 3 \times \cos 60^\circ = 90 \times 3 \times 0.5 = 135 \text{ J}$. (Option B, 270 J, forgets the $\cos\theta$ factor; option D uses $\sin 60^\circ$ instead of $\cos 60^\circ$.)

SLO: U6 §6.1 (Vector Representation, Fig. 6.2), P-09-B-61 ($W = Fd$, force at an angle).

Q24. Renewable energy resource: → **D – Geothermal heat** **1 mark**

Natural gas, uranium and petroleum are all non-renewable (fossil/nuclear fuels). Geothermal heat, though extracted faster than it regenerates deep in the Earth, is described in the textbook as "practically renewable" given the Earth's enormous size.

SLO: U6 §6.3.6, P-09-B-70 (differentiate and list renewable/non-renewable resources).

Q25. Motor lifts 200 kg through 6 m in 10 s. Power: → **C – 1200 W** **1 mark**

Work done $W = mgh = 200 \times 10 \times 6 = 12,000 \text{ J}$. Power $P = W/t = 12,000 / 10 = 1200 \text{ W}$.

SLO: U6, P-09-B-73 (define and calculate power).

Q26. Fuel used in nuclear power plants: → **B – Uranium-235** **1 mark**

The textbook states nuclear power plants typically use a rare type of uranium called U-235 (with Plutonium-239 as another usable fuel produced within reactors).

SLO: U6 §6.3.2 Nuclear Fuel, P-09-B-71 / P-09-B-72.

Q27. Physics textbook's Key Points define technology as: → **D – Tools/methods that help people accomplish tasks** **1 mark**

This is the Key Points' exact definition of technology. Option A describes science; option B swaps in the book's definition of engineering; option C is the definition of a hypothesis.

SLO: U9 §9.4, P-09-G-08 (differentiate science, technology, engineering).

Q28. Branch of physics studying Earth's climate: → **A – Climate Physics** **1 mark**

The textbook defines Climate Physics as "the branch of physics that deals with the study of the Earth's climate," distinct from Geophysics (Earth's structure/processes) and Acoustics (sound).

SLO: U9 §9.2 (branches of physics), P-09-G-02.

Q29. Best definition of a 'hypothesis': → **C – A tentative explanation made before research, later verified** **1 mark**

This is the textbook's own definition of a hypothesis. Option A describes a law; option B describes a theory; option D describes an untested/unfalsifiable idea, not a hypothesis.

SLO: U9 §9.3, P-09-G-06 (understand hypothesis, theory and law).

Q30. Physics is a branch of: → **D – Physical Sciences** **1 mark**

Science is broadly divided into Physical Sciences (Physics, Chemistry, Geology, Geography, etc.) and Biological Sciences (Botany, Zoology, Medicine, etc.). Physics belongs to the Physical Sciences.

SLO: U9 introduction; Exercise Q8; P-09-G-03 (physics as a subset of physical/natural sciences).

SECTION B — Short Questions (2 × 4 = 8 Marks)

2(i) Theory vs Law, with two examples of each **4 marks**

Theory: A theory explains *how* nature behaves under specific conditions — it is a well-supported explanation of some aspect of the natural world, built up from a hypothesis that has survived repeated testing. **1 mark**

Examples: **Kinetic Molecular Theory; Theory of General Relativity.** **1 mark** (0.5 each)

Law: A law is a statement that summarises an observed regularity or pattern in nature and gives the mathematical relationship between the variables involved — it describes *what* happens for a narrower set of conditions. **1 mark**

Examples: **Newton's Laws of Motion; Newton's Law of Universal Gravitation.** **1 mark** (0.5 each)

Key difference (for reference): laws tend to describe a narrower relationship between two specific quantities; theories are more expansive and focus on explaining the how and why of a phenomenon.

2(ii) Weight on Earth and on the Moon **4 marks**

(a) Weight on Earth: $W = mg = 55 \times 9.8 = 539 \text{ N}$. **1.5 marks** (formula 0.5, substitution & answer with unit 1)

(b) Weight on the Moon: $W = mg = 55 \times 1.62 = 89.1 \text{ N}$. **1.5 marks** (formula 0.5, substitution & answer with unit 1)

(c) The **mass does not change** — it stays 55 kg. Mass is the quantity of matter in a body and does not depend on gravity; only the **weight** changes, because it depends on the gravitational field strength, which is different (smaller) on the Moon. **1 mark**

SECTION C — Long Question (1 × 8 = 8 Marks)

3. Orbital motion of a satellite 8 marks

(a) **Orbital motion** is the regular, repeating path that one object in space (a satellite) takes around another, more massive object (e.g. Earth), held in place by gravity. 1 mark Average orbital speed formula: $v = 2\pi r / T$, where r = orbital radius and T = orbital period (time for one revolution, and distance travelled in one orbit = $2\pi r$, the circumference). 1 mark

(b) Given $r = 6800 \text{ km} = 6.8 \times 10^6 \text{ m}$, $T = 5440 \text{ s}$: 0.5 mark

$v = 2\pi r / T = (2 \times 3.1416 \times 6.8 \times 10^6) / 5440$ 1 mark

$v = 42,725,760 / 5440 = 7854 \text{ m/s}$ ($\approx 7.85 \times 10^3 \text{ m/s}$, or 7.85 km/s) 0.5 mark

(c) Converting to km/h: $v = 7854 \times 3.6 = 28,274.4 \text{ km/h}$ ($\approx 2.83 \times 10^4 \text{ km/h}$). 1 mark

(d) The centripetal force is provided by the Earth's **gravitational force (gravity)** acting on the satellite. Its direction is always **towards the centre of the Earth** (the centre of the circular orbit). 1 mark

(e) Given $m = 750 \text{ kg}$, $v = 7854 \text{ m/s}$, $r = 6.8 \times 10^6 \text{ m}$: 0.5 mark

$F = mv^2/r = 750 \times (7854)^2 / (6.8 \times 10^6)$ 1 mark

$F = 750 \times 61,685,316 / 6,800,000 \approx 6803 \text{ N}$ ($\approx 6.80 \times 10^3 \text{ N}$) 0.5 mark

— End of Answer Key —